

Tonii: Customizable Amulets that Activate Augmented Reality Suits

Yingxi Adelle Lin

adelle@touchtech.io

Touch Interactive Technologies

Brooklyn, NY, USA

Matthew Pinner

matt@touchtech.io

Touch Interactive Technologies

Brooklyn, NY, USA

ABSTRACT

This work introduces Tonii, an integration of augmented reality (AR) technology, artistic expression, community engagement and social empowerment. Tonii empowers participants to craft unique amulets adorned with symbols that encapsulate fundamental values for cultivating safe and inclusive digital realms. Utilizing advanced machine learning techniques, these symbols serve as keys to unlock AR suits, offering users a tangible connection between virtual experiences and real-world identities. This endeavor not only narrows the divide between digital and tangible realms but also underscores the critical roles of technological inclusivity and societal empowerment.

CCS CONCEPTS

• Human-centered computing → Mixed / augmented reality.

KEYWORDS

Amulet, Wearable, Machine Learning, Spatial Web

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1 INTRODUCTION

Tonii is an ongoing research and art initiative introducing wearable AR technology for personal and collective expression. Participants craft custom amulets that, when detected, unveil virtual suits in augmented reality, representing values ranging from self-awareness to information accessibility. Informed by a wide survey of empowering social dynamics, these values emphasize emotional support, authenticity, and trust in digital communities. The project addresses critical themes of tech equity, access, literacy, justice, and inclusion, particularly in the spatial web. Through activating digital extensions with wearable amulets, Tonii encourages participants to engage with their selves across virtual and physical realms, advocating for collective accountability, future-oriented ancestorship, and just virtual environments.

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2 DESIGN PHILOSOPHY AND TECHNOLOGY

A virtual avatar study categorizes users into Realistics, Ideals, Roleplayers, and Fantasists. The design of the Tonii suits reinforces the imaginative desires of Roleplayers and Fantasists, seeking to transcend their realities [Neustaedter and Fedorovskaya 2009]. AR distinguishes itself from fully virtual avatars by offering the capability to amalgamate the virtual with participants' physical presence, creating a newly integrated self. This raises the question: How can we design virtual embodiments to foster care, stewardship, and support, enriching users' sense of self and community engagement?

Toni Cade Bambara's concept of "sistering" emphasizes empowerment, community building, and collective action [Brown 2019]. This project draws on these concepts, infusing them with digital technology for transformative experiences. The project evolves through co-creation with participants, gathering input through surveys, workshops, and live installations (Figure 1).

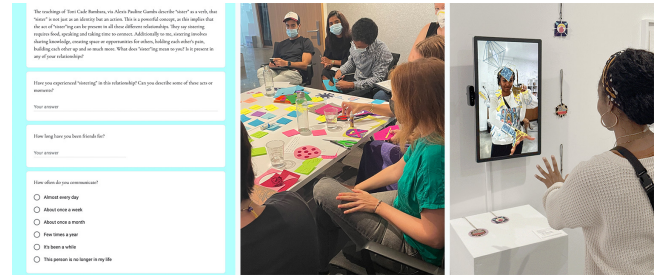


Figure 1: (a) Tonii online survey (b) Tonii workshop conducted with Nour Batyne (c) Gallery installation

According to the Feminist Principles of the Internet, "The internet is a space where social norms are negotiated, performed, and imposed, often in an extension of other spaces shaped by patriarchy and heteronormativity. Our struggle for a feminist internet is one that forms part of a continuum of our resistance in other spaces, public, and private." [FPI 2014] The project challenges digital norms, promoting a vision of the internet free from heteronormative biases. By equipping communities with tools for meaningful technology engagement, the project enhances technological equity, literacy, and accessibility.

3 IMPLEMENTATION

3.1 Participant Created Amulets

For the installation, participants are invited to construct their own amulets (Figure 2a) from a selection of symbols (Figure 2b) that represent values and themes that they feel should exist in spatial web communities. The values are generated from workshops and

surveys conducted over the last 2 years [Lin and Batyne 2022] and the symbols themselves are rooted in a rich tapestry of cultural symbolologies such as Underground Railroad Quilt codes, Hobo codes, Adinkra symbols, Tarot symbols and other cultural representations. The symbols are available to the participants as printed laser cut materials that they can assemble on-site into a single amulet. The goal is to empower participants to articulate their visions for a more inclusive and diverse digital ecosystem.

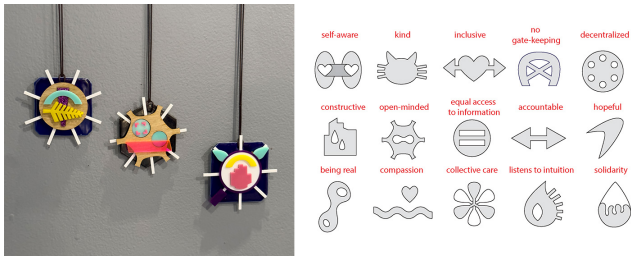


Figure 2: (a) Completed amulets by artist (b) Subset of values symbology

3.2 AR technology

Once assembled, participants can present the amulets to a camera at the installation. An object recognition model, trained with a set of symbols, identifies these symbols and correlates them to the AR outcome. Each symbol corresponds to a part of the virtual garment or texture, resulting in a unique virtual suit tailored to the chosen values. The project leverages AR technology that is able to recognize the participants body as a mesh and fit custom 3D modeled parts (Figure 3) on the body like a suit, allowing participants to inhabit these suits and move as if they are actually wearing them, facilitating a deeply immersive experience. The machine learning model is integrated into the AR software in order to create the full project capabilities.

3.3 AR and social impact

This project explores AR's potential as a tool for social empowerment and community, addressing what Nancy Fraser identified, that



Figure 3: (a) Render of AR suit 3D model (b) Render of AR suit 3D model

marginalized groups are excluded from a universal public sphere [Fraser 1990]. As AR / Virtual Reality (VR) / Extended Reality (XR) worlds are becoming the new public squares, it is important to include voices of marginalized groups in these worlds. By weaving the narratives of underserved communities into the fabric of AR/VR/XR spaces, Tonii aims to transform these platforms into inclusive forums for those underserved voices.

3.4 Take home and participant experience

The initiative offers participants a chance to continue their engagement beyond the physical installation, enabling them to experience the virtual suits via an AR-enabled application. This extended interaction promotes ongoing dialogue around themes of sistering, accountability, and collective values, highlighting the project's dedication to democratizing technology.

4 OBSERVATIONS AND DISCUSSION

Observations from prior public installations in galleries reveal a notable transformation in participants' body language, transitioning from serious expressions to expressions of lightness and joy upon the manifestation of the suits on their bodies. Research highlights the pivotal role of avatar appearance in identity representation and user behavior modulation within virtual environments [Neustaedter and Fedorovskaya 2009], a principle extending to AR. Participants became more conversational and eager to share their experiences and insights. This phenomenon underscores technology's potential to democratize art and personalize empowerment experiences through digital augmentation.

In summary, Tonii amalgamates art, technology, and social activism in the evolving digital ecosystem. Leveraging AR and spatial web technologies, it introduces a unique mode of engagement and empowerment, enriching discussions on technological equity, inclusivity, and future digital realms. Committed to sisterhood, accountability, and communal values, Tonii establishes a foundational model for integrating art and technology to promote inclusive and empowering virtual spaces.

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